

Manthan Bhala

Machine Learning Engineer & Software Developer

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EXPERIENCE

Senior Machine Learning Engineer — Makro PRO

Oct 2023 – Present

Remote

- Developed user-based recommendations using collaborative filtering and two-tower models, and product-based recommendations using collaborative filtering and vector embeddings.
- Built vector search for e-commerce by fine-tuning LLM embeddings for semantic product retrieval.
- Built personalisation models through e-commerce-specific feature engineering and LLM distillation into LightGBM for production-scale inference.
- Developed end-to-end recommendation pipeline with retrieval, ranking, and reranking stages.
- Improved search results through autocategorization, autocorrect, query expansion, and query rewriting models.
- Built a chatbot POC with RAG pipeline and function calling, optimized with quantization and vLLM deployment.

Machine Learning Engineer — Frinks AI

Mar 2023 – Jul 2023

Bengaluru, Karnataka, India

- Developed and optimized ML inference pipelines for real-time defect detection and counting on industrial production lines.
- Built computer vision and backend services for a quality inspection platform for dimensioning, surface defect, and crack detection.

Data Scientist — Jaguar Land Rover

Sep 2020 – Feb 2023

Bengaluru, Karnataka, India

- Developed and maintained ML pipelines and backend services for Vehicle Algorithmic Self Diagnostics Tool.
- Worked on Advanced Driver Assistance Systems (ADAS) for autonomous and assisted driving features.

SKILLS

ML Engineering

Python · PyTorch · TensorFlow

Applied AI

LLMs · Recommendation Systems · Information Retrieval

Data & Infrastructure

SQL · Docker · Kubernetes

EDUCATION

Bachelor of Technology — Indian Institute of Technology (IIT), Bombay

2016 – 2020 | CGPA: 8.48/10

PATENTS

2501007631 — A Hybrid Lexical-Semantic Retrieval Process for E-Commerce Search

2501004476 — A Hybrid LLM Process for Multi-Objective Reranking in E-Commerce

2501004419 — A Multilingual Phonetic BK-Tree Based Spelling Correction Process in E-Commerce